2. Vehicle must stop immediately after applying the brake.
3. Brake shoes and brake linings should have better anti-wear property.
4. It should not be over sensitive.
5. It must be durable and have less maintenance.
6. The operation of the braking system should not interfere the operation of the other systems of the vehicle.
7. Braking system should not be noisy during its operating condition.

## 6.3 Types of brake

The brakes are classified into a number of types. They are,

### i. According to the application

1. Normal brake or foot brake
2. Hand brake or parking brake

### ii. According to application of brake shoe

a. Drum type brake
   1. Internally expanding type brake
   2. Externally contracting type brake
b. Disc type or caliper type brake

### iii. According to the power

1. Mechanical brake
2. Hydraulic brake
3. Air brake
4. Vacuum brake and
5. Electric brake

### 6.3.1 Mechanical brake

Mechanical brakes are classified into two different types, they are shown as follow

1. Internal Expanding Brake
2. External Contracting Brake

#### Internally Expanding Brake

In this braking system the brake shoes are mounted inside of brake drum and the lower ends of brake shoes are connected with help of anchor pins and brake carrier plate. Based on this the movement of brake shoe is positioned. In between the upper ends of brake shoes one brake cam is mounted. Brake pedal is located at the right side corner of the driver's leg. When the brake pedal is pressed, a long rod which is connected with the brake cams in the four brake carrier plate simultaneously are expanded inside the brake drum which causes friction that stops the rotating wheel. When the brake pedal is released, the brake cam is brought to its original position. The brake shoes are also brought to their original positions at the same time without touching the brake drum due to the spring's compressive force. Hence the wheels on the brake drum rotate immediately without any friction.

#### Externally contracting type brake

This type of braking system is mostly used in cars as hand brake. In this system the outer layer of the brake drum has a brake lining which is contact with the brake band. The brake band is operated by brake lever and the linkages to stop the rotating brake drum. By pulling the hand brake lever, the brake band is forced to press the brake drum tightly. Hence the rotation of brake drum is stopped and the vehicle is stopped. When the hand lever is released, the brake band



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